STATS 200A Lecture 10

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Definition 10.1 (Conditioning on Information)
For $X \in L^2$

$$\mathbb{E}[X \mid \mathcal{G}] = \underset{Z \in L^2(\mathcal{G})}{\arg\min} \mathbb{E}(X - Z)^2$$

In particular, $\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)]$

Definition 10.2 (Orthogonality)
$\mathbb{E}[RW] = 0, \quad \forall W \in L^2(\mathcal{G})$

Remark 10.3
For $X \in L^1, M = \mathbb{E}[X \mid \mathcal{G}]$ is the unique almost surely variable satisfying

$$M \in L^1, M \text{ is } \mathcal{G} - \text{measurable}, \qquad \mathbb{E}[M\mathbf{1}_A] = \mathbb{E}[X\mathbf{1}_A] \quad \forall A \in \mathcal{G}$$

Remark 10.4 (Conditioning on an Observation)

$$\mathbb{E}[x \mid Y] = \mathbb{E}[X \mid \sigma(y)] = g^\star(Y).$$

For the die, $g^\star(1) = 3$ and $g^\star(0) = 4$.

$$g^\star = \underset{g:g(Y)\in L^2}{\arg\min} \|X - g(Y)\|_2 = \underset{g:g(Y)\in L^2}{\arg\min} \mathbb{E}(X - g(Y))^2$$

$(X_1, Y_1), \ldots, (X_n, Y_n) \overset{iid}{\sim} (X, Y)$

$\mathbb{P}_{(X_1,Y_1),\ldots,(X_n,Y_n)} \sim \mathbb{P}_{(X,Y)}^{\bigotimes n}$

$Z_1, \ldots, Z_n \overset{iid}{\sim} L, \qquad \frac{1}{n}\sum_{i=1}^n Z_i \xrightarrow{P} \mathbb{E}(Z_i)$

Linearity	$\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$
Monotonicity	$X \le Y \Rightarrow \mathbb{E}[X \mid \mathcal{G}] \le \mathbb{E}[Y \mid \mathcal{G}]$
Total expectation	$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$
Already known	X is $\mathcal{G}$-measurable $\Rightarrow \mathbb{E}[X \mid \mathcal{G}] = X$
Independent information	X is independent of $\mathcal{G} \Rightarrow \mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$

Remark 10.5 (Pull out what is known)

$$\mathbb{E}[WX \mid \mathcal{G}] = W\mathbb{E}[X \mid \mathcal{G}], \qquad (W \text{ bounded}, \mathcal{G} - \text{measurable})$$

Theorem 10.6 (Tower Property)

For $X \in L^1$ and $\mathcal{H} \subseteq \mathcal{G}$,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$$

Theorem 10.7 (Explained Variance + Remaining Prediction Error)

$$\mathrm{Var}(X) = \mathrm{Var}(M) + \mathbb{E}R^2$$